

Virenpuri Vasantpuri Vairagi

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LinkedIn | GitHub | Portfolio | Live ML App

Professional Summary

Aspiring AI Engineer and Data Scientist dedicated to building scalable, automated machine learning pipelines and intelligent software solutions. Highly proficient in Python, Scikit-learn, and the full development lifecycle—from raw data preprocessing to deploying real-time applications via Streamlit. Experienced in integrating AI with IoT and LLM-powered automation workflows to solve complex operational challenges. Proven researcher and problem-solver committed to delivering robust, data-driven systems that drive business efficiency.

Education

S.P.B Patel Engineering College

Mehsana, Gujarat

Bachelor of Engineering in Computer Engineering (CGPA: 8.00)

2022 – 2026

Experience

Brainy Beam Info-Tech Pvt. Ltd.

Ahmedabad, Gujarat

Data Science & Machine Learning Intern

01/2026 – 03/2026

- Architected the complete end-to-end Data Science pipeline to develop a highly accurate job offer acceptance prediction system, drastically improving human resources forecasting capabilities.
- Processed and encoded target variables, specifically mapping 'Joined' to 1 and 'Not Joined' to 0, ensuring precise model evaluation and pristine training data integrity.
- Evaluated multiple classification algorithms comprehensively, conclusively determining that Random Forest outperformed alternatives such as Gradient Boosting in precision and reliability.
- Deployed the finalized machine learning model to a live Streamlit Cloud environment, empowering recruitment teams to conduct immediate, real-time applicant analysis.

InfoLabz IT Services Pvt. Ltd.

Ahmedabad, Gujarat

Data Analysis & Machine Learning Intern

07/2025 – 07/2025

- Constructed clean, automated data pipelines utilizing Python, Pandas, and Scikit-learn to streamline extensive exploratory data analysis (EDA).
- Executed rigorous data preprocessing protocols that significantly enhanced the readiness, accuracy, and reliability of datasets for supervised learning evaluations.

Projects

Job Offer Acceptance Prediction

Technologies: Python, Pandas, Scikit-Learn, Random Forest, Streamlit

- Formulated an advanced machine learning solution to forecast candidate dropouts, equipping technical recruiters with actionable, data-driven predictive insights.

- Refined raw salary expectation data through meticulous cleaning and EDA, extracting critical behavioral indicators that strengthened the model's overall predictive power.
- Engineered and launched an interactive web application via Streamlit Cloud, utilizing the optimized Random Forest algorithm to deliver seamless, real-time forecasting results to end-users.

AI-Powered Customer Feedback Classification

Technologies: n8n, Google Gemini, Airtable, APIs

- Built a production-ready automation workflow combining n8n and Google Gemini to seamlessly capture, parse, and analyze unstructured customer feedback.
- Categorized incoming data streams dynamically into actionable segments (Complaints, Compliments, Feature Requests), significantly reducing manual review time.
- Orchestrated continuous database synchronization and automated team notifications utilizing Airtable and Slack APIs to accelerate internal response times.

Adaptive & Collaborative Traffic Control System

Technologies: AI/IoT, Random Forest, V2X, Streamlit

- Integrated real-time IoT sensor data with machine learning frameworks to accurately monitor traffic density and predict urban congestion patterns.
- Designed a collaborative decision-making platform leveraging V2X communication to automate emergency vehicle prioritization, ensuring highly efficient path-clearing for ambulances.
- Implemented an adaptive traffic signal control mechanism by feeding continuous IoT data streams directly into a Random Forest classifier to achieve dynamic system responsiveness.

Publications & Extracurricular

- **Publication:** Published peer-reviewed research titled “Multi-feature Collaborative Traffic Management: An AI and IoT-Integrated Approach” in the 5th International Conference on Advanced Computational and Communication Paradigms, Springer (DOI: 10.1007/978-3-032-13544-5_39), February 2026.
- **Hackathon:** Awarded 3rd Prize at the Smart India Hackathon (SIH) 2025 out of numerous competitive entries by prototyping an innovative Smart Traffic Management System aimed at mitigating urban congestion.

Technical Skills

- **Languages & Databases:** Python, SQL, HTML5, CSS3, JavaScript, PHP, Java, C
- **Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Streamlit
- **AI & Automation:** LLM Integration, n8n, Model Context Protocol (MCP)
- **Software & Tools:** n8n, PyCharm, Jupyter Notebook, VS Code, Power BI, Airtable, Tableau, GitHub/Git, XAMPP